

- Attendance (Add names below)
 - Sharabh
 - Joseph
 - Thomas
 - Anam
- Note: Meetings are Thursday 6-7 at Pub Aff 2238 and Friday 6-7 at Hedrick Study; required to make at least 1 a week for checkins, Fridays will usually be workshops/presentations
- SDXL, ControlNet, IP-Adapter Presentation
 - Add questions if needed
- Add name if you were able to set up GitHub + clone the repo
 - Zijian
 - Anam
- Setting up Colab
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- Voting on site design
 - We're using Bauhaus
- Panel JSON object:
 - Brainstorming:
 - Spec recommends shot_type, camera_angle, characters, lighting_mood, background, and action_note
 - Final object:
 - {
 - shot_type: "" # enum
 - camera_angle: "" # enum
 - characters: "" #
 - lighting_mood: ""
 - background: ""
 - action_note: ""
- Assignments
 - All
 - Think about what attributes need to be added to the Panel JSON
 - Test out Colab code (Make sure to activate T4 x 2 GPU before running; Settings -> Accelerator):
<https://www.kaggle.com/code/sharabhojha/storyboard-gen-experiment>
 - Thomas
 - OpenRouter LLM; pick a free LLM api that can descriptively yet consistently output a Panel JSON object
 - Nemotron 3 Nano 30B A3B (free): Pros: Did a perfect job at adding additional details and produced correct json format consistently.

- Joseph
 - Tune ControlNet parameters and document findings
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- Anam
 - Tune IP-Adapter parameters and document findings
 - With Human reference (used Zendaya):
 - Low temp (~0.3)
 - Hero looked like young kid, expression was slanted eyes with slight blush and line for mouth
 - No hero/villain present for 1st or 2nd scenes
 - Scene detail/coloring more coherent compared to other lvls
 - Med temp (~0.5)
 - Hero looked like a guy, skin is lighter, eyes/expression is gone
 - No hero/villain present for 1st or 2nd scenes
 - High temp (~0.7)
 - Hero looked like a guy, one eye is missing
 - No hero/villain present for 1st or 2nd scenes
 - With Non-Human reference (used Luffy and Elmo):
 - Low temp (~0.3)
 - Luffy results:
 - More detailed
 - Didn't look like elmo
 - Character was a boy with light skin and black spiky hair
 - Some character present in scene 1 and scene 3, not that consistent across
 - Scene 1 character had blond hair and red/blue body
 - Scene 3 character had black hair and brown/blue body
 - Elmo results:
 - Just colored the default scenes with elmo's color scheme
 - No added detail or character traits other than colors
 - Med temp (~0.5)
 - Luffy results:
 - Mixed up coloring of the character's hair/clothing
 - Character build is more muscular than the reference
 - Expression is only two lines for eyes and then a dot for the mouth
 - Colors of reference matched for scene 1 but not scenes 2 or 3

- Elmo results:
 - Full body kinda looks like a teletubby
 - No character in scene 3
 - Just colored the default scenes with elmo's color scheme
 - No added detail or character traits other than colors
- High temp (~0.7)
 - Luffy results:
 - Scene 1 same as 0.5, Scene 3 better (matches ref colors more)
 - Scene 2 too muscular, right side of body is irregular and character has no expression
 - Elmo results:
 - Each scene has much less detail than other temperatures
- Overall Conclusions
 - Low temp better for Human reference
 - Low/Medium temp better for Nonhuman reference
 - **THOMAS: Add an IP-Adapter temperature attribute to the panel jsons based on this**

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- Attendance:
 - Thomas
 - Nigella

5/1

- Attendance:
 - Anam
 - Joseph
- Assignments
 - Anam, Joseph: Integrate and stress test current image generation with controlnet openpose
 - Sharabh, Nigella: Semantic search on the openpose dataset, retrieve most relevant poses, modify reference images to fit ideal size/camera angle before passing to openpose
 - Thomas: Build openrouter script to parse visual style, sentence description, n panels and return list of panel jsons; start working on fastapi
 - **OpenRouter key:**
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

Integrate and stress test current image generation with ControlNet OpenPose.

```
ip_adapter_scale = 0.3
controlnet_scale = 0.6
inference_steps = 25
guidance_scale = 7.5
image_size = (1024, 1024)
```



```
ip_adapter_scale = 0.4
controlnet_scale = 0.6
inference_steps = 25
guidance_scale = 7.5
image_size = (1024, 1024)
```



```
ip_adapter_scale = 0.5
controlnet_scale = 0.6
inference_steps = 25
guidance_scale = 7.5
image_size = (1024, 1024)
```



```

ip_adapter_scale = 0.6
controlnet_scale = 0.6
inference steps = 25
guidance scale = 7.5
image size = (1024, 1024)

```



```

ip_adapter_scale = 0.7
controlnet_scale = 0.6
inference steps = 25
guidance scale = 7.5
image size = (1024, 1024)

```



Depends on the result that `ip_adapter_scale = 0.4`, change the `controlnet_scale`.

```

ip_adapter_scale = 0.4
controlnet_scale = 0.5
inference steps = 25
guidance scale = 7.5
image size = (1024, 1024)

```



```

ip_adapter_scale = 0.4
controlnet_scale = 0.6
inference steps = 25
guidance scale = 7.5
image size = (1024, 1024)

```



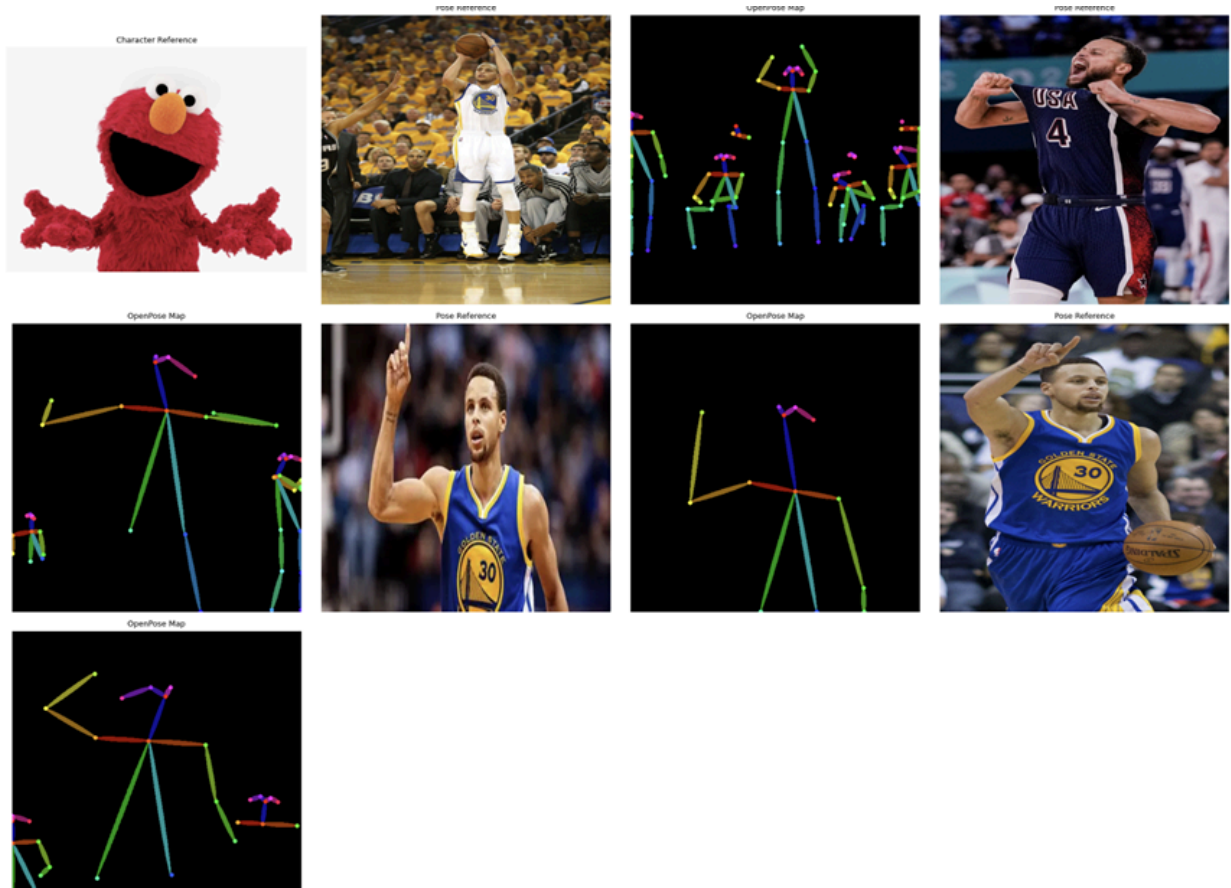

```
ip_adapter_scale = 0.4
controlnet_scale = 0.7
inference steps = 25
guidance scale = 7.5
image size = (1024, 1024)
```



```
ip_adapter_scale = 0.4
controlnet_scale = 0.8
inference steps = 25
guidance scale = 7.5
image size = (1024, 1024)
```



Results of ControlNet OpenPose by changing parameters.



Reference photos and skeleton.

`Ip_adapter_scale`: Increase this parameter will make output closer to the reference photo.

`ControlNet_scale`: Increase this parameter will make output closer to the skeleton generated by OpenPose.

Overall, open pose controlNet works for the result.